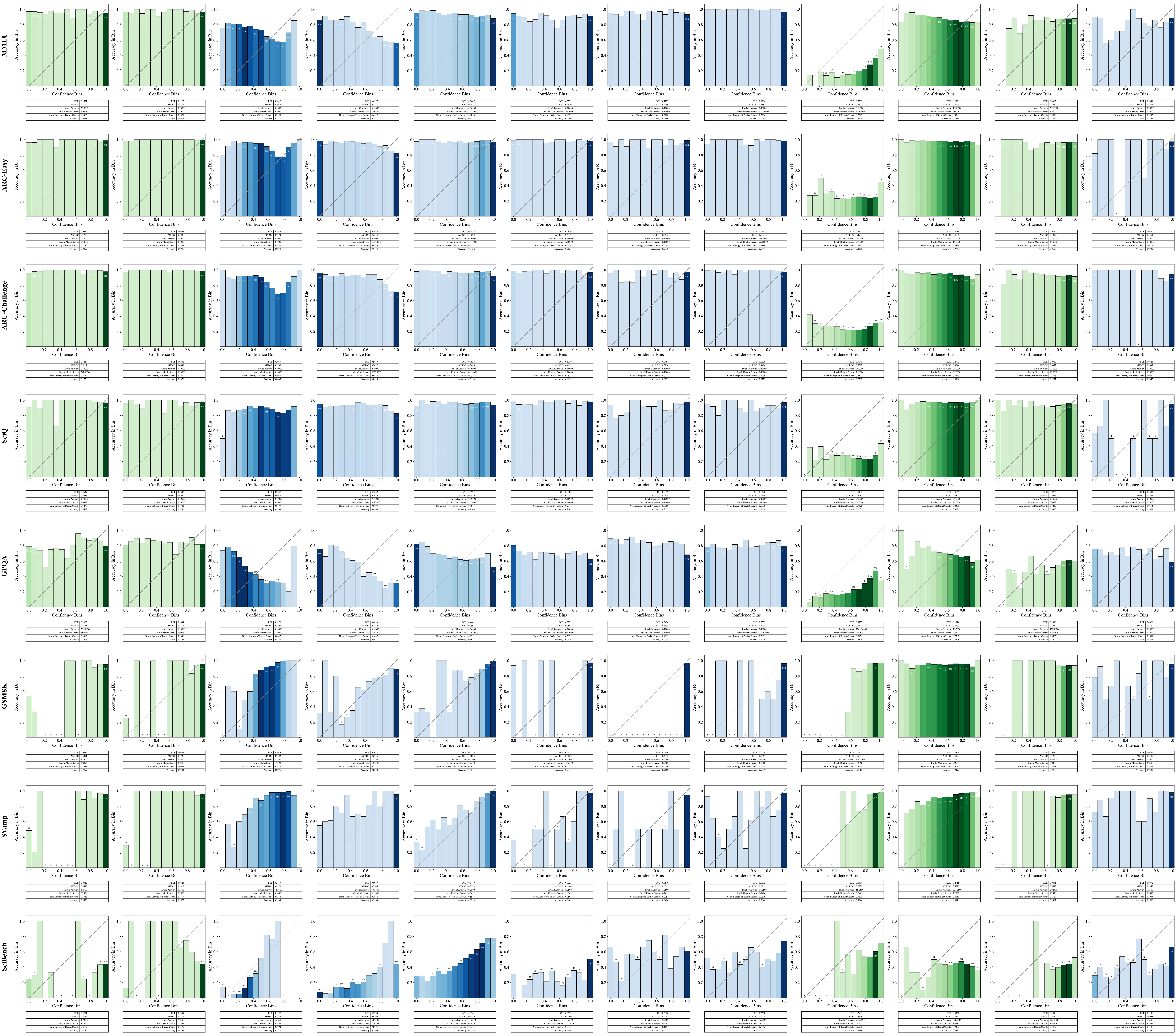
Calibration Plots for Uncertainty Metric P(True)



MMLU	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.1551	0.1225	0.3422	0.6277	0.3667	0.3729	0.1139	0.1394	0.5261	0.2368	0.0682	0.1491
AUROC	0.4892	0.5337	0.3985	0.3151	0.3837	0.4510	0.4459	0.4182	0.6177	0.4303	0.5069	0.5483
Invalid Answers	2/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	18/10000	29/10000	0/10000
Invalid Metric Scores	2/9998	0/10000	0/10000	374/10000	121/10000	20/10000	0/10000	375/10000	12/10000	48/9982	64/9971	0/10000
Norm. Entropy of Bucket Counts	0.3065	0.3071	0.9190	0.6117	0.8858	0.4321	0.2180	0.3206	0.7245	0.8589	0.4979	0.1453
Accuracy	0.9578	0.9684	0.7218	0.7458	0.9232	0.9200	0.9360	0.9738	0.2499	0.8651	0.8779	0.8861

ARC-Easy	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0675	0.0554	0.2924	0.5443	0.1435	0.0926	0.0231	0.0331	0.5257	0.3188	0.0726	0.0280
AUROC	0.5780	0.5086	0.3774	0.2644	0.4628	0.4742	0.5636	0.5074	0.4989	0.4606	0.5222	0.5364
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid Metric Scores	0/10000	0/10000	0/10000	618/10000	34/10000	3/10000	0/10000	22/10000	2/10000	1/10000	1/10000	0/10000
Norm. Entropy of Bucket Counts	0.1735	0.1883	0.8780	0.6780	0.6788	0.2014	0.0587	0.1119	0.7232	0.8431	0.4837	0.0117
Accuracy	0.9862	0.9936	0.8938	0.9144	0.9752	0.9824	0.9878	0.9844	0.2498	0.9748	0.9653	0.9732

ARC-Challenge	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.1224	0.0781	0.3059	0.5879	0.1922	0.1310	0.0467	0.0620	0.5040	0.2992	0.0720	0.0621
AUROC	0.5117	0.4688	0.3700	0.2657	0.3686	0.4633	0.5156	0.4494	0.5392	0.4244	0.4972	0.5037
Invalid Answers	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000
Invalid Metric Scores	242/10000	0/10000	0/10000	520/10000	42/10000	2/10000	0/10000	8/10000	1/10000	0/10000	1/10000	0/10000
Norm. Entropy of Bucket Counts	0.2554	0.2289	0.8940	0.6487	0.7373	0.2524	0.0927	0.1436	0.7547	0.8530	0.5618	0.0289
Accuracy	0.9770	0.9878	0.8405	0.8528	0.9514	0.9691	0.9713	0.9729	0.2500	0.9394	0.9277	0.9445

SciQ	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0637	0.0605	0.2637	0.5069	0.1922	0.0988	0.0359	0.0428	0.5188	0.3165	0.0765	0.0501
AUROC	0.5032	0.4960	0.4617	0.3470	0.4426	0.5181	0.5819	0.5319	0.5014	0.4855	0.5392	0.5364
Invalid Answers	1/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	2/10000	0/10000
Invalid Metric Scores	1/9999	0/10000	0/10000	927/10000	32/10000	4/10000	0/10000	0/10000	0/10000	2/10000	0/9998	0/10000
Norm. Entropy of Bucket Counts	0.1278	0.1481	0.8677	0.6693	0.7813	0.2171	0.0780	0.0670	0.7328	0.8396	0.5217	0.0114
Accuracy	0.9658	0.9756	0.8809	0.8968	0.9549	0.9762	0.9752	0.9674	0.2501	0.9686	0.9520	0.9501

GPQA	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.2665	0.2494	0.3176	0.6813	0.4469	0.5273	0.3434	0.3503	0.3733	0.1814	0.2999	0.4628
AUROC	0.4651	0.4813	0.3205	0.2779	0.3598	0.3885	0.3638	0.4937	0.6397	0.4503	0.5242	0.3869
Invalid Answers	266/10000	6/10000	0/10000	0/10000	0/10000	0/10000	0/10000	0/10000	1243/10000	78/5000	430/10000	0/10000
Invalid Metric Scores	85/9734	0/9994	1/10000	382/10000	272/10000	90/10000	3/10000	2250/10000	405/8757	40/4922	179/9570	3/10000
Norm. Entropy of Bucket Counts	0.3012	0.2648	0.8697	0.5087	0.8391	0.5597	0.4011	0.5805	0.8956	0.7587	0.4505	0.3801
Accuracy	0.8016	0.8193	0.5123	0.5678	0.6678	0.7019	0.7234	0.7970	0.2526	0.6598	0.5999	0.6304

GSM8K	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0547	0.0502	0.3085	0.1022	0.0226	0.0333	0.0340	0.0400	0.0471	0.3541	0.0446	0.0899
AUROC	0.6091	0.5726	0.7622	0.6346	0.8006	0.5022	0.4942	0.5363	0.5554	0.5029	0.4908	0.6587
Invalid Answers	4/2500	2/2500	8/2500	12/2500	0/2500	0/2500	0/2500	0/2500	134/2500	0/2500	12/2500	0/2500
Invalid Metric Scores	1/2496	0/2498	1/2492	33/2488	0/2500	18/2500	0/2500	6/2500	0/2366	2/2500	0/2488	2/2500
Norm. Entropy of Bucket Counts	0.1043	0.0644	0.7417	0.3137	0.4618	0.0229	0.0000	0.0268	0.3676	0.9094	0.3851	0.1374
Accuracy	0.9459	0.9484	0.8976	0.8582	0.9544	0.9734	0.9660	0.9599	0.9611	0.9492	0.9373	0.9432

SVamp	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.0442	0.0423	0.2657	0.0967	0.0410	0.0333	0.0570	0.0372	0.0434	0.2911	0.0451	0.0892
AUROC	0.5688	0.5813	0.8170	0.7146	0.8978	0.6560	0.6012	0.6761	0.6830	0.6725	0.5479	0.7943
Invalid Answers	4/2500	0/2500	54/2500	39/2500	9/2500	8/2500	1/2500	14/2500	55/2500	201/2500	24/2500	1/2500
Invalid Metric Scores	0/2496	0/2500	0/2446	81/2461	0/2491	25/2492	0/2499	0/2486	0/2445	2/2299	1/2476	0/2499
Norm. Entropy of Bucket Counts	0.1428	0.1169	0.8358	0.2644	0.5548	0.0567	0.0254	0.0876	0.3639	0.8650	0.3327	0.2075
Accuracy	0.9559	0.9572	0.9309	0.9164	0.9398	0.9603	0.9400	0.9646	0.9644	0.9316	0.9487	0.9560

SciBench	gpt-oss-20b	gpt-oss-120b	Ministral-8B	Mistral-Nemo-7B	Mistral-Small-3.2-24B	Llama-3.3-70B	Llama-4-Scout-17B	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-Llama-70B	DeepSeek-Qwen-32B	google/gemma-3-27b
ECE	0.5201	0.5339	0.1297	0.2268	0.1302	0.4524	0.3920	0.2600	0.2993	0.3351	0.4490	0.3190
AUROC	0.5797	0.5112	0.7348	0.6860	0.6859	0.5700	0.6061	0.6644	0.5759	0.4926	0.5270	0.6933
Invalid Answers	45/2500	9/2500	55/2500	60/2500	11/2500	14/2500	9/2500	12/2500	743/2500	328/2500	291/2500	6/2500
Invalid Metric Scores	8/2455	1/2491	0/2445	129/2440	0/2489	111/2486	0/2491	10/2488	5/1757	83/2172	34/2209	4/2494
Norm. Entropy of Bucket Counts	0.2112	0.1375	0.6094	0.8705	0.9364	0.3443	0.1651	0.6042	0.5258	0.7309	0.4387	0.4586
Accuracy	0.4140	0.4257	0.1820	0.1809	0.5195	0.4644	0.6078	0.6473	0.5919	0.4385	0.4317	0.5486